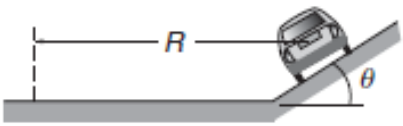
PHY101: Introduction to Physics-1

Monsoon, 2018

Tutorial-5

1. A skier leaves the ramp of a ski jump with a velocity of 10 m/s, 15.0° above the horizontal, as in the figure. The slope is inclined at 50.0°, and air resistance is negligible. Find (a) the distance from the ramp to where the jumper lands and (b) the velocity components just before the landing.

2. A car enters a turn whose radius is R, The road is banked at angle , and the coefficient of friction between the wheels and the road is . Find the maximum and minimum speeds of the car to stay on the road without skidding the sideways.



3. A cannon fires successively two shells with velocity v_o = 250 m/s; the first at the angle θ_1 = 60° and the second at the angle θ_2 = 45° to the horizontal, the azimuth being the same. Neglecting the air drag, find the time interval between firings leading to the collision of the shells.

4. A small piece of Styrofoam packing material is dropped from a height of 2.00 m above the ground. Until it reaches terminal speed, the magnitude of its acceleration is given by $a = g - bv$. After falling 0.500 m, the Styrofoam effectively reaches terminal speed, and then takes 5.00 s more to reach the ground. (a) What is the value of the constant b? (b) What is the acceleration at $t = 0$? (c) What is the acceleration when the speed is 0.150 m/s?